Report

Introduction

The assignment concerns the evaluation of the currency pair time series data. The objective of this evaluation is to understand the trend in volatility and white noise of the currency over the period.

As we are considering currency exchange pair, factors responsible for change in their values must be understood. This would help understand the significance of volatility and white noise. The exchange rate could be seen as "price of currency" which depends on its "demand and supply". Mainly two factors determine this said demand and supply.

1. Purchasing power parity which entails increasing price of domestic currency due to increase in demand of domestic goods by foreign entities which are bought in domestic currency resulting in increased value of the domestic currency.
2. Covered interest rate parity by this we mean countries with higher interest rate would attract foreign investors in hopes of higher return resulting in increased value of the domestic currency. As currency is regulated by a central body, which if does not increase the currency in circulation, increases its value.

Its also important to understand the concepts of white noise and volatility.

White Noise: The sudden changes in the time series i.e., high frequency changes. Mathematically, White noise is defined as "A generalized stationary stochastic process with constant spectral density". Another definition for it is "A time series is called white noise if it is a sequence of independent and identically distributed random variable with finite mean and variance". This in financial terms means random collection of variables that are uncorrelated. The presence of absence of any given phenomenon has no relationship with any other.

Volatility: The lower frequency change (i.e., the smoother change in time series) are considered to be volatility of signal. Volatility in finance is generally regarded as a metric that measures the magnitude of the change in prices.

Data Acquisition and Cleaning

Financial data is acquired though yfinance library, which provides the data from Yahoo! Finance. It's Ticker() module facilitates acquiring market data for the financial instrument. Following parameters were chosen for evaluation.

Currency Pair	**Symbol**	**Period**	**Data Collected**
Euro-US Dollar	EURUSD	01/01/2011-01/01/2021	Close Price
Great Britain Pound - Japanese Yen	GBPJPY	01/01/2011-01/01/2021	Close Price
Swiss Franc – Chinese Yaun	CHFCNY	01/01/2011-01/01/2021	Close Price

Other parameters like Opening, High, Low Price and Volume, Dividends etc. though available, were dropped as Closing values would suffice for the task. Data (Close Price) for all currency pair thus obtained was merged and stored in pandas DataFrame data structure.

Missing Values

In the initial *sanity check* it was observed for the period, certain values were missing for the currency pairs. This could have resulted in discrepancies between the pairs and a proper comparison between them would not have been possible.

To avoid this, missing values were estimated to be the arithmetic mean of previous and next day price. It maintains the trend in price and thus was considered the best choice. The resulting time series for various pairs could be seen below.

Initial and final twenty values were removed from the dataset to avoid variation. Removing the initial values does not affect the overall trend in a large dataset as this and resent values could be removed as we are more concerned about the overall trend of the currencies.

Data Decomposition

Before moving on to the decomposition of data, attention must be given in the choice of decomposition method used.

The time series data seen above could be seen having both slowing changing tends with some sudden change. As we are concerned with the variation or trend of the entire time series, the amplitude of frequency (change in value over time) for both general trend and abrupt changes are important. Here, method like Fourier transform is not sufficient as when moving from time domain to frequency domain we lose the time data. This is especially important for our evaluation as it is relevant to understand the changes in historic perspective. It is also not efficient to register abrupt changes. Wavelet transform on the other has gives us both the frequency amplitude and time localization.

Wavelets

Wavelets is rapidly changing wavelike oscillation which has zero mean and exist for a finite duration. It has two main properties i.e.

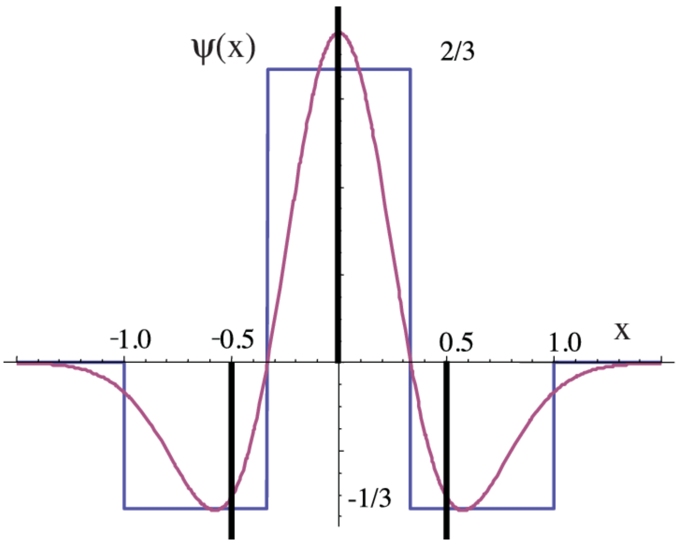
1. **Scaling** which is refers to the process of stretching or shrinking the signal in time. Scaling factor (s) is a positive value and is inversely proportional to frequency. Large scale factor results in stretched wavelet and corresponds to lower frequency which captures the slow changes in time series. Small scale factor results in shrunk wavelet and corresponds to higher frequencies and captures abrupt changes.
2. **Shifting** is delaying or advancing start of the wavelet along the length of signal to align wavelet with the desired feature to capture it.

These properties are to be chosen correctly to capture the required change in time series data.

Wavelet transformation in mainly of two type of wavelet transform. Continuous Wavelet Transform or CWT and Discrete Wavelet Transform or DWT. For our application we shall be using the CWT as it is the more ideal candidate for signal or time series analysis due to its more fine-grained resolution and is usually chosen in most tasks (e.g., singularity detection). *

For this the PyWavelets library is used and the original time signal is decomposed into seven layers using the proper parameters

Wavelet Transform	**Scales**	**Mother wavelet**
CWT	1-7	Mexican Hat Wavelet [mexh]

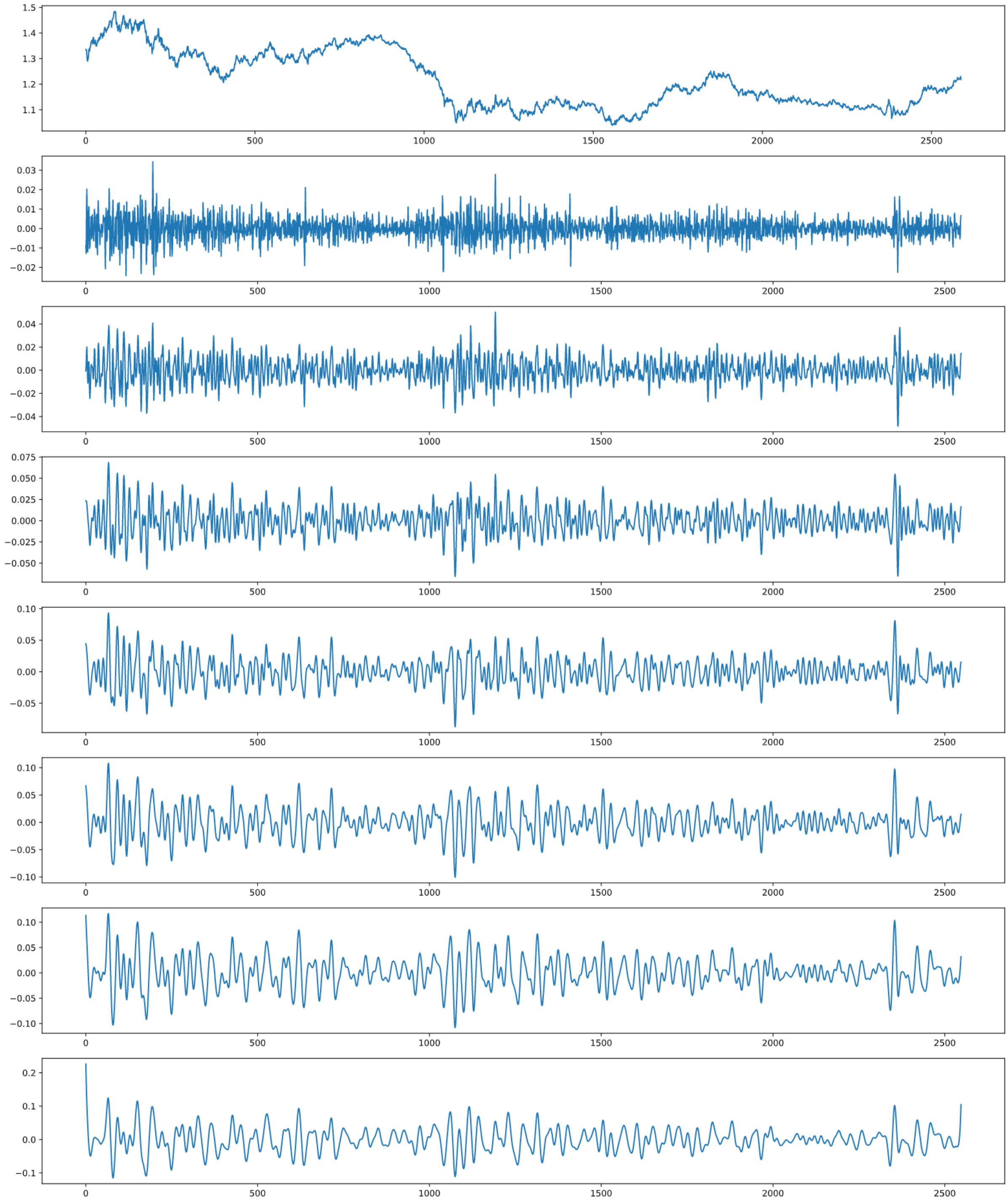


The Mexican hat is mathematically expressed as below.

$$\psi(t) = \frac{2}{\sqrt{3}\sqrt[4]{\pi}} \exp^{-\frac{t^2}{2}} \left(1 - t^2\right)$$

This results in the decomposition of the timeseries data in seven layers. Only the decomposition of Euro-US dollar pair has been shown here.

* [Using continuous verses discrete wavelet transform in digital applications - Signal Processing Stack Exchange]

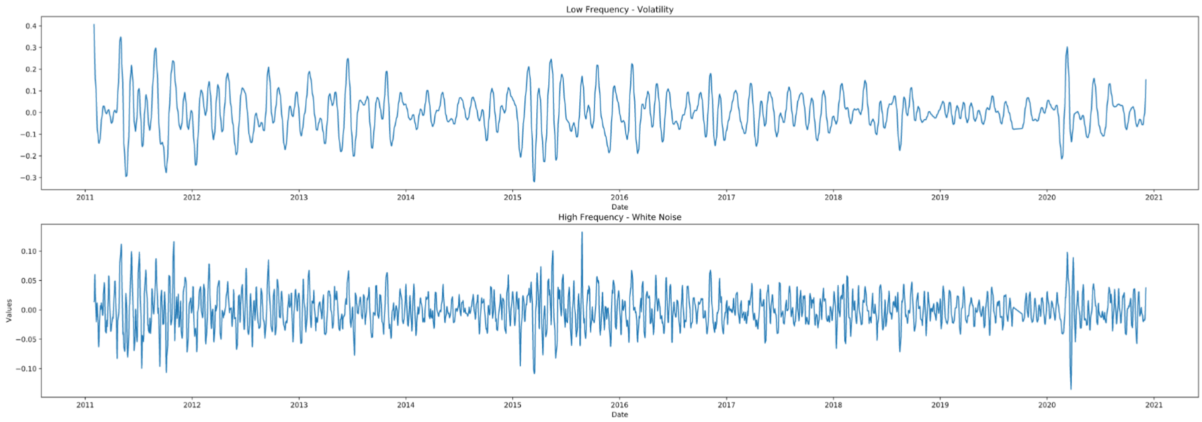


Data Compression

As could be seen above the data in decomposed into 7 layers each layer with capturing different frequency of the original signal. We can broadly compress the data into high frequency and low frequency and look at their trends individually. The high frequency corresponds to the white noise and the lower frequency corresponds to the volatility of the time series. Compression can be achieved using various methods like SVD, weighed average, simple summation etc. Here two methods are discussed namely summation and SVD.

Summation

This could be considered the most rudimentary compression method where top three i.e., first three high frequency layers are summed together to form a resulting high frequency layer and Similarly, bottom three i.e., last three low frequency layer are summed together to for high frequency layer.



Singular Value Decomposition (SVD)

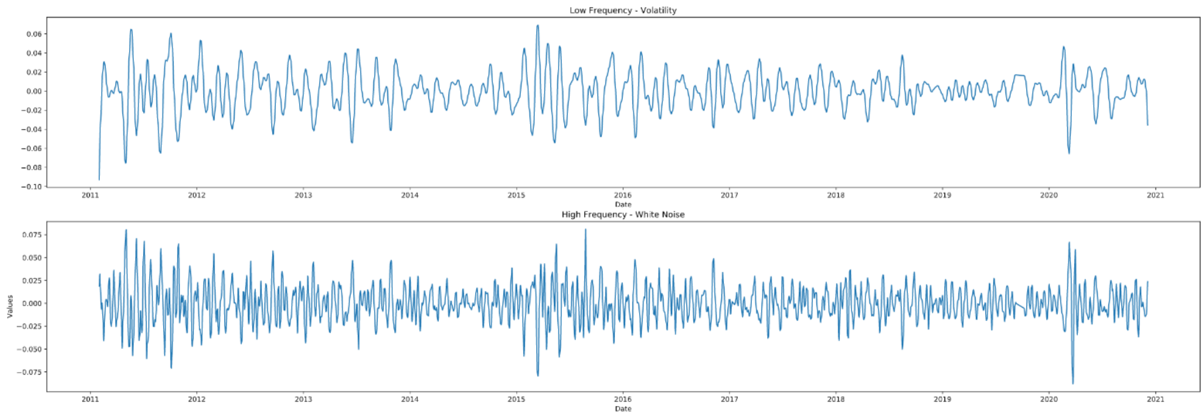
The second method used for compression was SVD, it could be represented mathematically as

$$\mathbf{A}_{nxp} = \mathbf{U}_{nxn}\, \mathbf{S}_{nxp}\, \mathbf{V}^{\mathbf{T}}{}_{pxp}$$

SVD intails that a matrix can be expressed as product of other three matrices. Where n represents number of rows and p represents number of dimentions . U represents matrix of left singular vectors in the columns, S the diagonal matrix with singular values and V matrix of right singular vectors in the columns.

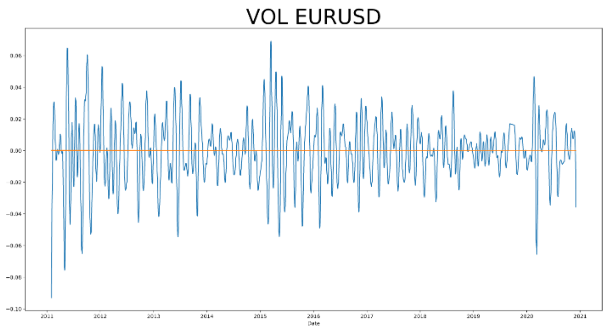
Compression with SVD takes advantage that very few singular values are large.

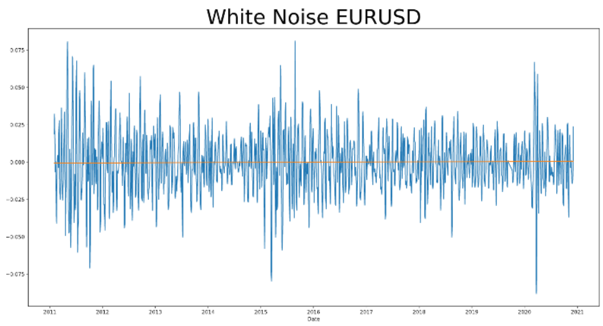
The result of compression is as follows.



Regression

To observe the general trend of the volatility and white noise we use linear regression. For this we have used the sklearn library from which Linear Regression model is used. Model has been fit to both forms of compression to compare the results from both compression methods. Here we only plot form one such combination (linear regression for SVD compression)





The slope of the line obtained form the regression is important to understand the general trend go volatility and white noise. The results of various combinations are discussed in the next section.

Conclusion

Volatility of all the currency pair irrespective of the compression method chosen is found to be negative. This correlates with the trend of decreasing volatility in the market. The magnitude of these slopes is found to be very small. Whereas with White Noice we see that compression method of SVD gives us positive slops with all currency pairs, but summation gives us negative slops for two pairs. This could be due to the error in compression method (summation).

Comparison between the results is given in the following table.

Currency Pair	Compression	Volatility	White Noice
EURUSD	SVD	R2: 4.497840955686172e-06 Slope: [[-5.72217323e-08]]	R2: 9.439891648654886e-05 Slope: [[2.62145542e-07]]
	Summation	R2: 8.979135745923461e-05 Slope: [-6.01397922e-07]	R2: 2.682255037855796e-06 Slope: [-3.20593456e-08]
CHFCNY	SVD	R2: 5.085274860094913e-06 Slope: [[-4.31480295e-08]]	R2: 5.602618938480795e-05 Slope: [[1.14845193e-07]]
	Summation	R2: 3.420458007441507e-05 Slope: [-3.91454252e-06]	R2: 3.528436407940205e-06 Slope: [-4.30794473e-07]
GBPJPY	SVD	R2: 3.96456900864095e-07 Slope: [[-1.20503698e-08]]	R2: 4.801719877267629e-08 Slope: [[4.19373405e-09]]
	Summation	R2: 3.122296333701868e-07 Slope: [-6.15723887e-06]	R2: 6.110381922663777e-08 Slope: [8.16361243e-07]